

SOLAR CELLS

Reducing nonradiative recombination in perovskite solar cells with a porous insulator contact

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Inserting an ultrathin low-conductivity interlayer between the absorber and transport layer has emerged as an important strategy for reducing surface recombination in the best perovskite solar cells. However, a challenge with this approach is a trade-off between the open-circuit voltage (V_{oc}) and the fill factor (FF). Here, we overcame this challenge by introducing a thick (about 100 nanometers) insulator layer with random nanoscale openings. We performed drift-diffusion simulations for cells with this porous insulator contact (PIC) and realized it using a solution process by controlling the growth mode of alumina nanoplates. Leveraging a PIC with an approximately 25% reduced contact area, we achieved an efficiency of up to 25.5% (certified steady-state efficiency 24.7%) in p-i-n devices. The product of $V_{oc} \times FF$ was 87.9% of the Shockley-Queisser limit. The surface recombination velocity at the p-type contact was reduced from 64.2 to 9.2 centimeters per second. The bulk recombination lifetime was increased from 1.2 to 6.0 microseconds because of improvements in the perovskite crystallinity. The improved wettability of the perovskite precursor solution allowed us to demonstrate a 23.3% efficient 1-square-centimeter p-i-n cell. We demonstrate here its broad applicability for different p-type contacts and perovskite compositions.

Advanced device architecture and contact design, together with absorber quality improvement, have driven the rise of perovskite solar cells (1). Power-conversion efficiencies (PCEs) as high as 25.7% have been achieved for conventional n-i-p devices with the perovskite deposited on the electron transport layer (ETL) (2, 3). The p-i-n cells [an inverted architecture built on the hole transport layer (HTL)] recently also reached certified PCEs of ~24% (4, 5). Although PCEs of p-i-n cells lag behind those of their n-i-p counterparts, they are highly valued, especially for their demonstrated greater stability and higher efficiency in state-of-the-art tandem cells (6–10). The open-circuit-voltages (V_{oc} s) of these top-performing cells have reached >90% relative to the Shockley-Queisser (S-Q) limit for the corresponding band gap, exceeding the level of record silicon cells (~85%) (1, 11). This high level of V_{oc} reflects the intensive efforts to reduce nonradiative recombination at photocarrier transport interfaces realized by the insertion of ultrathin low-conductivity passivation layers. Low-dimensional perovskites (4, 12–17) and low-conductivity organic materials [such as polystyrene, poly(methyl methacrylate) polymer (PMMA), and organometallic molecules] (3, 5, 18, 19) are the dominant passivation materials for current top-performing cells. Researchers have also explored dielectric inorga-

nycs such as Al_2O_3 and ZrO_2 (20–24), analogous to the tunnel oxide passivation in silicon cells (25–28). However, they are far less successful, primarily because of the extreme sensitivity of electric tunneling to the dielectric thickness. It has been reported that a subnanometer increase in the dielectric layer thickness can cause a vast fill factor (FF) reduction because of the series resistance loss (20, 26). This trade-off problem is also generic for a two-dimensional (2D) perovskite and insulating organic passivation layers. This passivation-transport trade-off makes the simultaneous maximization of V_{oc} and FF a challenge (1, 18, 19). It also makes up-scaling fabrication a further challenge because it is not easy, especially using a solution process, to cover a rough large-area surface uniformly with an ultrathin passivation layer.

Here, we present a contact structure to overcome these two challenges in which the conventional ultrathin (~1 nm) passivation layer is replaced with a thick (~100 nm) dielectric mask that has random nanoscale openings (Fig. 1A). Photocarrier extraction is not compromised because photocarriers travel across the thick dielectric mask through the openings rather than by electrical tunneling. This contact design is called a “porous insulator contact” (PIC). We realized this design using a low-cost and scalable solution process that exhibits high tolerance to variations in dielectric thickness and opening size. Our numerical simulations and device experiment revealed that PIC reduces nonradiative recombination at perovskite’s HTL contact through a combined effect of contact area reduction and passivation. Moreover, because PIC clusters are hydrophilic and rough and the preferred HTL materials are hydrophobic, surface wet-

ting and perovskite film quality were improved. The surface recombination velocity (SRV) at the HTL contact was reduced from 64.2 to 9.2 cm/s. The bulk recombination lifetime was increased from 1.2 to 6.0 μ s because of improvements in the perovskite crystallinity. Using a PIC that can reduce the contact area by ~25%, we boosted the PCE of a p-i-n device from ~23 to 25.5% (certified steady-state PCE of 24.7%, active area 0.06 cm²), the highest ever reported for a p-i-n cell (table S1). Its product of $V_{oc} \times FF$ is 87.9% of the S-Q limit, rivaling that of the record n-i-p devices. PIC’s advanced scalability also allows us to demonstrate 23.3% PCE for a 1-cm² p-i-n cell, among the highest for all perovskite cells with active area ≥ 1 cm² (table S2). We demonstrated its broad applicability for different HTLs and perovskite compositions (band gaps of 1.55, 1.57, and 1.65 eV).

PIC design

The PIC design is inspired by but different from the local contact in commercial silicon solar cells (27–31). For instance, the dielectric passivation layer (Al_2O_3/SiN_x or SiO_2/SiN_x stack with oxide thickness of ~10 nm) in a passivated emitter and rear contact silicon cell is opened by laser firing to localize the metal contact (or heavily doped contact) for carrier extraction (fig. S1A). The typical opening area ratio is ~1% to minimize contact area for recombination and to retain a high FF. The spacing between openings matches the photocarrier diffusion length in silicon on a scale of multiple hundreds of micrometers to one millimeter. Unfortunately, this technique cannot be directly transferred to perovskite solar cells because the photocarrier diffusion length is on a scale of hundreds of nanometers. This nanoscale requirement leads to a fabrication challenge for laser firing or opening etching. For n-i-p perovskite solar cells, nanoscale local contact has been attempted by a periodic TiO_2 nanorod array (pitch of ~300 nm) fabricated using high-resolution electron-beam lithography (18). The pinheads of TiO_2 nanorods puncture the ultrathin (<3 nm) PMMA insulating passivation layer to form the local contacts with the perovskite for electron extraction. However, the sophisticated electron-beam lithography microfabrication process makes it difficult for mass production. The nanorod local contact had enhanced FF up to 84.5%, but also showed a slightly reduced V_{oc} because of the increased perovskite/nanorod contact area for recombination (fig. S1B).

In light of these considerations, we sought a PIC design (Fig. 1A and fig. S1C) for p-i-n devices with combined features of (i) a reduced contact area at the perovskite/HTL interface; (ii) opening spacing on a scale of ~100 nm, shorter than the photocarrier diffusion length of perovskite; and (iii) a scalable fabrication

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process. As illustrated in Fig. 1A, process *a* is the lateral transport of carriers that cannot electrically tunnel through the thick Al_2O_3 . Process *b* represents the “through-the-opening” transport of carriers through the channels formed by the infilling perovskite. We selected Al_2O_3 to be the dielectric material for its easy accessibility, but in principle, other insulating materials such as SiO_2 , ZrO_2 , and insulating organic dielectrics could also be used.

To disentangle the PIC's structural effect and interface passivation effect, we developed a 2D numerical drift-diffusion model (for details, see the materials and methods, fig. S2, and tables S3 and S4) (18, 32) to simulate devices with different dielectric coverages and different surface recombination velocities at the perovskite/HTL interface (tuning the defect density while the defect capture ability remained unchanged). The local insulator was modeled with thickness $h = 100$ nm to eliminate the electrical injection or tunneling through the dielectric layer. The local insulator's width $d = 100$ nm is shorter than the photocarriers' diffusion length in the perovskite to ensure their lateral transport between adjacent openings (18, 31). The percentage reduction of the contact area is tuned through the dielectric coverage $[d/(d+s)]^2$, where s is the local opening width. We started by applying PIC in a control device (bare HTL) with a PCE of ~22%, a respectable value for p-i-n cells. As shown in Fig. 1B, we found that the PCE, V_{oc} , and FF all increased steadily with the dielectric coverage, i.e., reduced the perovskite/HTL contact area, which agrees well with previous reports (26–28, 33). For PIC devices with different trap densities at the perovskite/HTL interface (fig. S3), we observed the same improvement trend of PCE, V_{oc} , and FF. This independence confirms the PIC's inherent structural effect on the V_{oc} and FF enhancement, providing a pathway to overcoming the passivation-transport trade-off encountered in conventional contacts. A PIC's coverage of 25% can deliver a respectable PCE percentage improvement of ~5% (Fig. 1, B and C, blue line). Increasing the PIC's coverage to 64% (Fig. 1C, orange dashed line) or coating this PIC on a passivated HTL with lower surface trap density (Fig. 1C, red line) can further increase the PCE percentage improvement to exceed 10%. These simulation results provide important guidance for achieving >25% efficient p-i-n solar cells. To verify the applicability of the PIC model, we also conducted simulations for varied insulator widths d (30, 100, and 200 nm) and higher coverage up to 81% (fig. S4), which delivered entirely consistent results. Beyond the perovskite/HTL contact area analyzed in the simulation, the interface passivation effect in real devices can also occur at the perovskite/dielectric contact interface. Therefore, it is rational to regard the above simula-

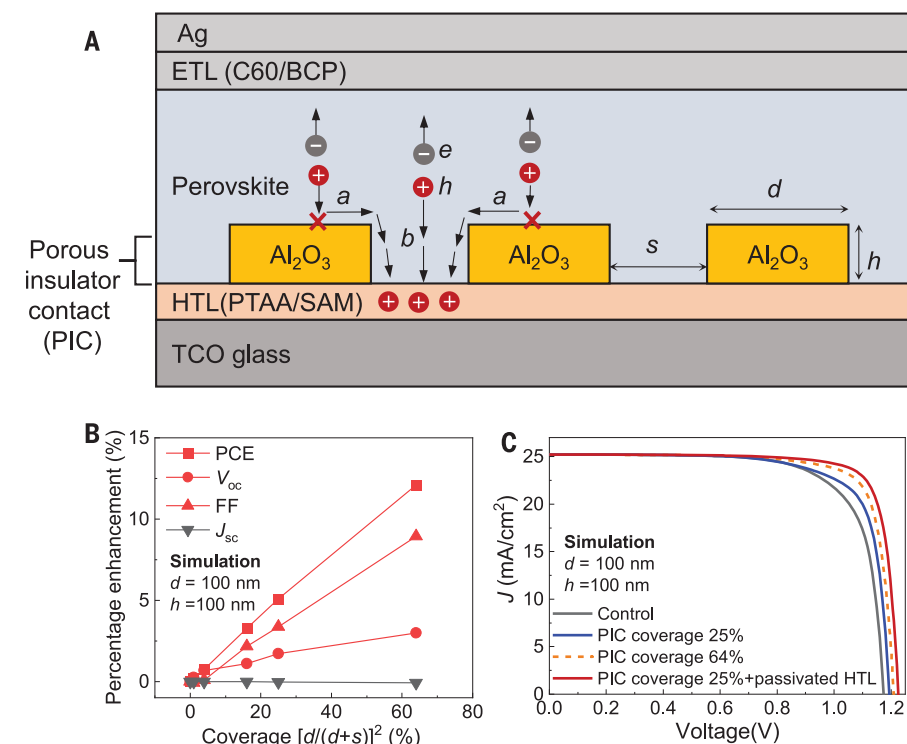


Fig. 1. Concept and simulation of a PIC for a p-i-n device. (A) Schematic description of the PIC device structure (not to scale). The parameters h , d , and s represent the local Al_2O_3 dielectric mask's height, width, and local opening width, respectively. Process *a* represents the lateral transport of carriers that cannot electrically tunnel through the thick Al_2O_3 . Process *b* represents the “through-the-opening” transport of carriers through the channels formed by the perovskite infill. The insertion of PIC reduces the perovskite/HTL contact area and suppresses interface recombination. The PIC coverage fraction is defined as $[d/(d+s)]^2$. e and h represent the electron and hole charge carrier, respectively. (B) 2D drift-diffusion simulation of a PIC device ($d = 100$ nm, $h = 100$ nm) built on a conventional HTL (such as a PTAA polymer) showing consistently increased PCE, V_{oc} , and FF as a function of PIC coverage fraction. The y axis shows their percentage enhancement relative to the control (bare HTL without PIC). (C) Simulated $J-V$ curves of the control device (without PIC, solid black line), the PIC devices with coverage of 25% (solid blue line) and 64% (orange dashed line) built on conventional HTL, and the PIC device with coverage of 25% built on a passivated HTL (reduced surface recombination velocity through reduced trap density at interface, solid red line). Simulation results [(B) and (C)] show that the PIC structural effect can be paired with HTL passivation to improve the photovoltaic performance.

tion as a conservative estimation. The PIC's reduced contact area, paired with the entire interface passivation, can gain even higher V_{oc} and FF.

PIC fabrication

We implemented the PIC contact design by judicious control over the growth mode of the Al_2O_3 nanoplate clusters during the solution process. The PIC principle requires a high contrast between the local dielectric mask and the adjacent opening area of HTL. Therefore, it is critical to avoid forming a continuous ultrathin dielectric layer that would create a passivation-transport trade-off. The former morphology desirable for PIC is analogous to the “isolated island” that results from Volmer-Weber (V-W) growth (Fig. 2E) (34), and the latter is analogous to the “layer-plus-island” morphology that arises from Stranski-Krastanov

(S-K) growth (Fig. 2D) (35). We first tested the 30-nm Al_2O_3 nanoplate dispersion ($d = 30$ nm) that has been used for mesoporous scaffolds (33). By spin-casting the dispersion with different concentrations (0.7 to 4 mg ml⁻¹), we found that its strong preference was to form continuous layers, as shown in atomic force microscopy (AFM) morphology images (Fig. 2, F to H). The Al_2O_3 coverage fraction rapidly increased to almost 100%, even at a diluted concentration of 2 mg ml⁻¹ (Fig. 2M). As per the S-K growth mechanism, we reason that this morphology occurs because the nanoplate-nanoplate interaction is weaker than the nanoplate-substrate interaction (35). The weak nanoplate-nanoplate interaction is suggested by the narrow peak of hydrodiameter size in dynamic light scattering (Fig. 2A) and the well-isolated dispersion manner shown in the transmission electron microscope (TEM) images

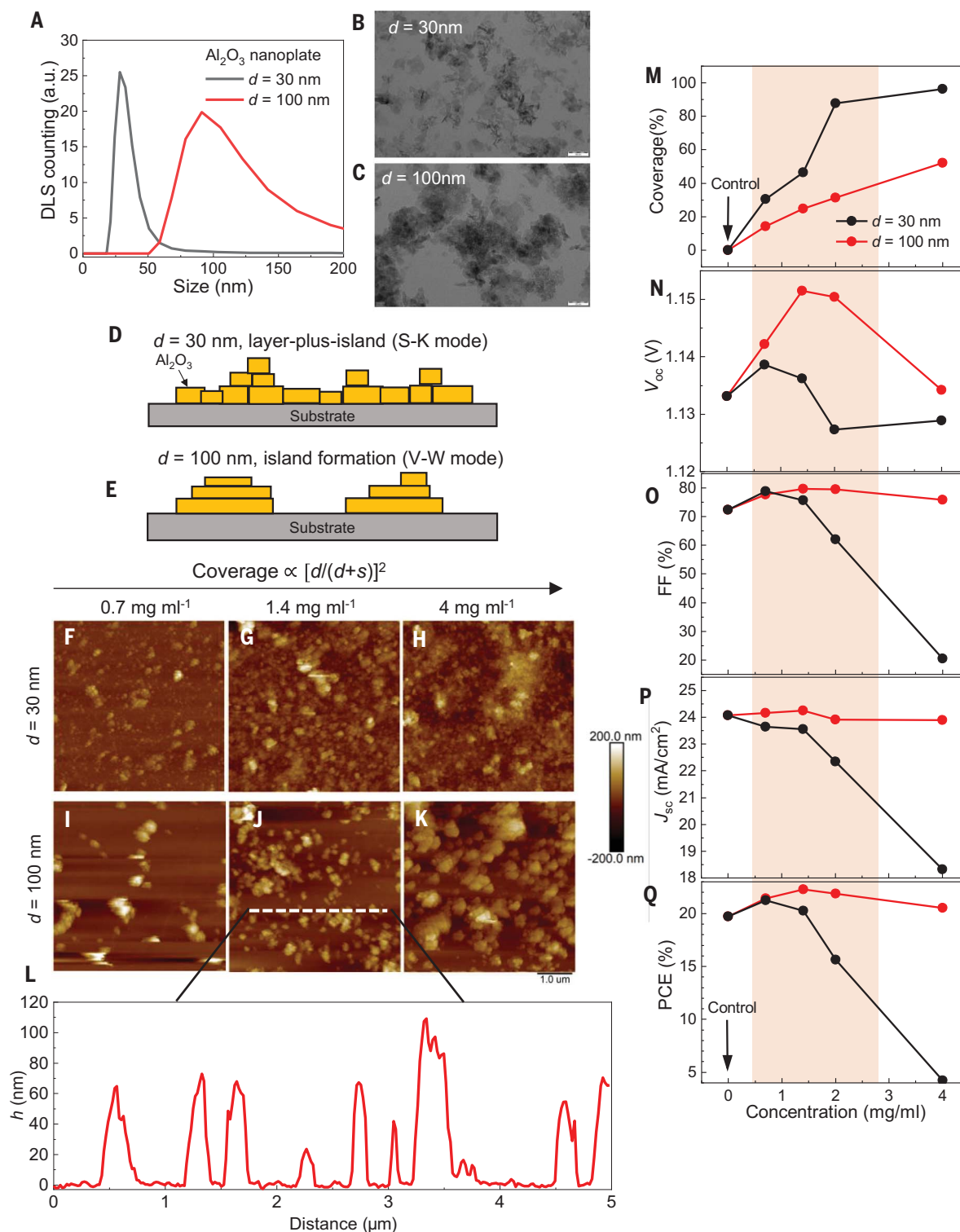


Fig. 2. Realization of PIC contact design through a solution process of Al_2O_3 nanoplate. (A) Dynamic light scattering (DLS) measurements of Al_2O_3 nanoplate in isopropanol (1 mg ml^{-1}) with two different hydrodiameter sizes ($d = 30 \text{ nm}$ and $d = 100 \text{ nm}$). (B and C) TEM images of Al_2O_3 dispersions ($\sim 0.1 \text{ mg ml}^{-1}$) with nanoplate size $d = 30 \text{ nm}$ (B) and $d = 100 \text{ nm}$ (C). (D) Illustration of S-K growth of a “layer-plus-island” using Al_2O_3 nanoplate dispersion with $d = 30 \text{ nm}$. (E) Schematic description of V-W growth of an “isolated island” using Al_2O_3 nanoplate dispersion with $d = 100 \text{ nm}$. (F to K) AFM images of ITO/HTL/ Al_2O_3 stacks using

$d = 30 \text{ nm}$ [(F) to (H)] and $d = 100 \text{ nm}$ [(I) to (K)] Al_2O_3 nanoplates with a 0.7 , 1.4 , and 4 mg ml^{-1} dispersion concentration, respectively. (L) Line scan from an AFM image of Al_2O_3 islands ($d = 100 \text{ nm}$, concentration = 1.4 mg ml^{-1}) after the V-W growth and coarsening procedure. The flat and clean baseline corresponds to the HTL surface at openings. (M to Q) Evolution of coverage fraction (M), V_{oc} (N), FF (O), J_{sc} (P), and PCE (Q) for two different Al_2O_3 sizes ($d = 30 \text{ nm}$ versus 100 nm) as a function of the dispersion concentration. The solar cell structure is ITO/PTAA/PIC/perovskite/C60/BCP/Ag. The HTL is the polymer PTAA.

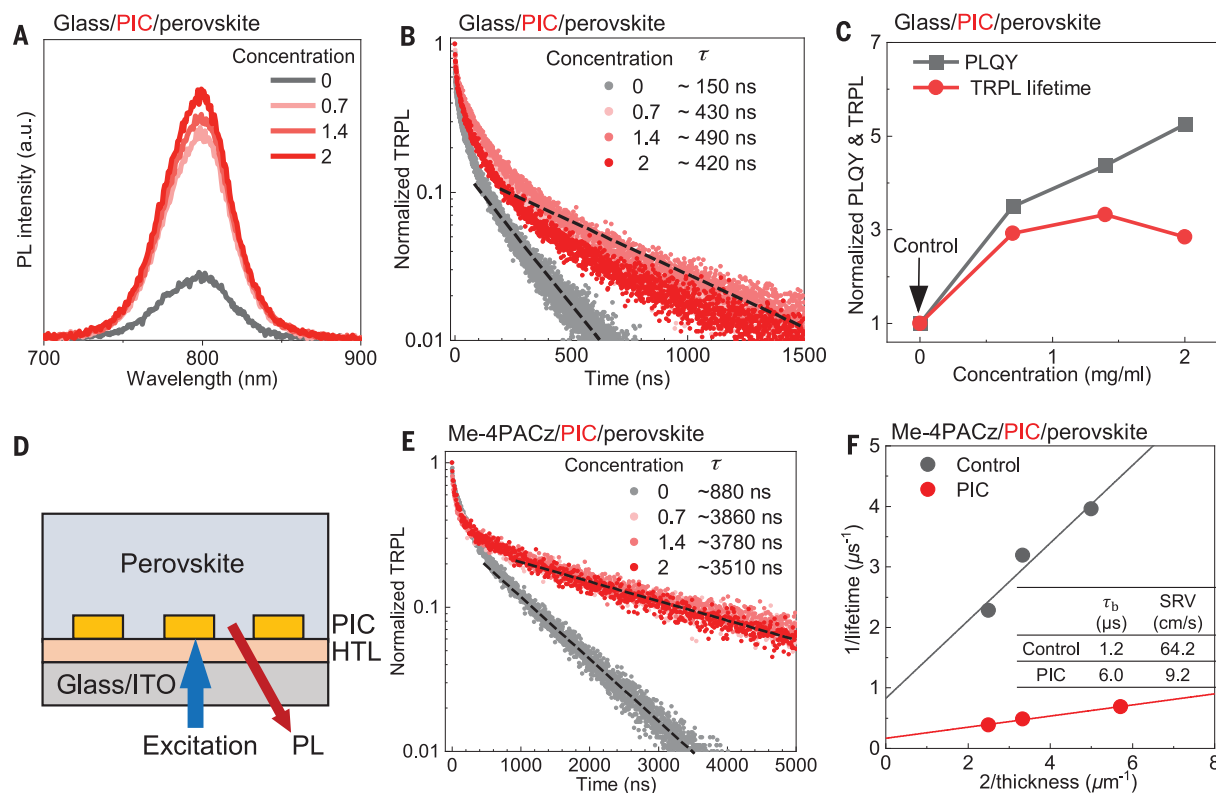


Fig. 3. Impact of PIC on reducing nonradiative recombination. (A and B) Steady-state photoluminescence (PL) spectra (A) and TRPL decays (B) of the glass/PIC/perovskite stacks. Al_2O_3 nanoplates ($d = 100$ nm) with concentrations of 0 to 2 mg ml^{-1} were used to form different coverage fractions. The slow monoexponential decay (τ , indicated by the black dashed line) represents the dominance of trap-assisted recombination in the bulk and the surface. The prolonged τ in PIC-containing samples indicates reduced non-radiative recombination. (C) Normalized PLQY and τ of glass/PIC/perovskite stacks (indicated as PIC sample) relative to the control sample (PIC

concentration = 0). (D) Schematic description of the PL/TRPL experiment on the ITO/HTL/PIC/perovskite stacks. (E) TRPL decays of ITO/HTL/PIC/perovskite stacks, where the HTL is Me-4PACz. Insertion of PIC at the HTL/perovskite interface can prolong the τ (indicated by the black dashed line) and enhance the PLQY, highlighting the reduced nonradiative recombination and the potential for V_{oc} and FF enhancement. (F) Substantially improved perovskite bulk lifetime (τ_b) and SRV in the PIC sample relative to the control revealed by fitting the TRPL transients of varied thickness through a double-sided heterostructure model.

(Fig. 2B). By contrast, the 100-nm Al_2O_3 nanoplates ($d = 100$ nm) formed islands and clusters in the fashion of V-W growth, as shown in AFM morphology images for the concentration of 0.7 to 4 mg ml^{-1} (Fig. 2, I to K). The broad peak of the hydrodiameter size shown in dynamic light scattering (Fig. 2A) and the clustering manner shown in TEM images (Fig. 2C) of the 100-nm Al_2O_3 suggest a strong nanoplate-nanoplate interaction, providing the driving force for the V-W growth. With increased concentration (0.7 to 4 mg ml^{-1}), we observed growth and coarsening of clusters, resulting in the coverage fraction gradually increasing from ~15 to 50% (Fig. 2M). A representative height profile at a concentration of 1.4 mg ml^{-1} (Fig. 2L) reflects the growth and coarsening of Al_2O_3 islands, with average island width of 160 nm (range 64 to 270 nm), average island height of 64.5 nm (range 25 to 109 nm), and coverage of ~25% (fig. S5). Crucially, the clean and flat background in morphology images, corresponding to the exposed HTL surface, can be attained for concentra-

tions ranging from 0.7 to 4 mg ml^{-1} (Fig. 2, I to K), providing the precondition for the successful realization of PIC.

Solar cell fabrication and testing

We then fabricated p-i-n devices to evaluate the solution-processed Al_2O_3 dielectric mask with different sizes ($d = 30$ versus 100 nm) and various concentrations (0.7 to 4 mg ml^{-1}) (Fig. 2, M to Q). Following the state-of-the-art p-i-n architecture (5, 7), we used poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) as the HTL deposited on indium tin oxide-coated glass (ITO), and C60 as the ETL deposited on top of the perovskite surface (for details, see the materials and methods). The perovskite composition was $\text{Cs}_{0.05}(\text{FA}_{0.95}\text{MA}_{0.05})_{0.95}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$ (where FA is formamidinium and MA is methyl ammonium) with a band gap of 1.55 eV. For the Al_2O_3 mask with $d = 30$ nm, we observed slightly increased V_{oc} and FF only for the concentration of 0.7 mg ml^{-1} . With further increased concentration, the FF decreased vastly, in agreement with the trend of short-

circuit current density (J_{sc}) loss and the formation of a continuous layer as observed in AFM images (Fig. 2, G and H). By contrast, we observed a significantly broader performance enhancement window for the Al_2O_3 mask with $d = 100$ nm. With increased concentration from 0.7 to 2 mg ml^{-1} , the coverage fraction was consistently increased. V_{oc} , FF, and PCE, as well as the retained J_{sc} , are fully subject to the prediction of the ideal PIC model and consistent with the V-W growth of islands. For the concentration of 4 mg ml^{-1} with an even higher coverage fraction (~50%), the V_{oc} and FF remained higher than the control device but less efficient than the trend extrapolation from the ideal PIC model. To explain this deviation at high concentration, we propose that the increased proportion of oversized islands (i.e., those larger than the photocarrier diffusion length of the perovskite) reduces carrier collection. Given the relatively large size dispersion in $d = 100$ nm solution, it is also possible that the S-K growth of smaller nanoplates occurs in parallel with V-W growth of larger

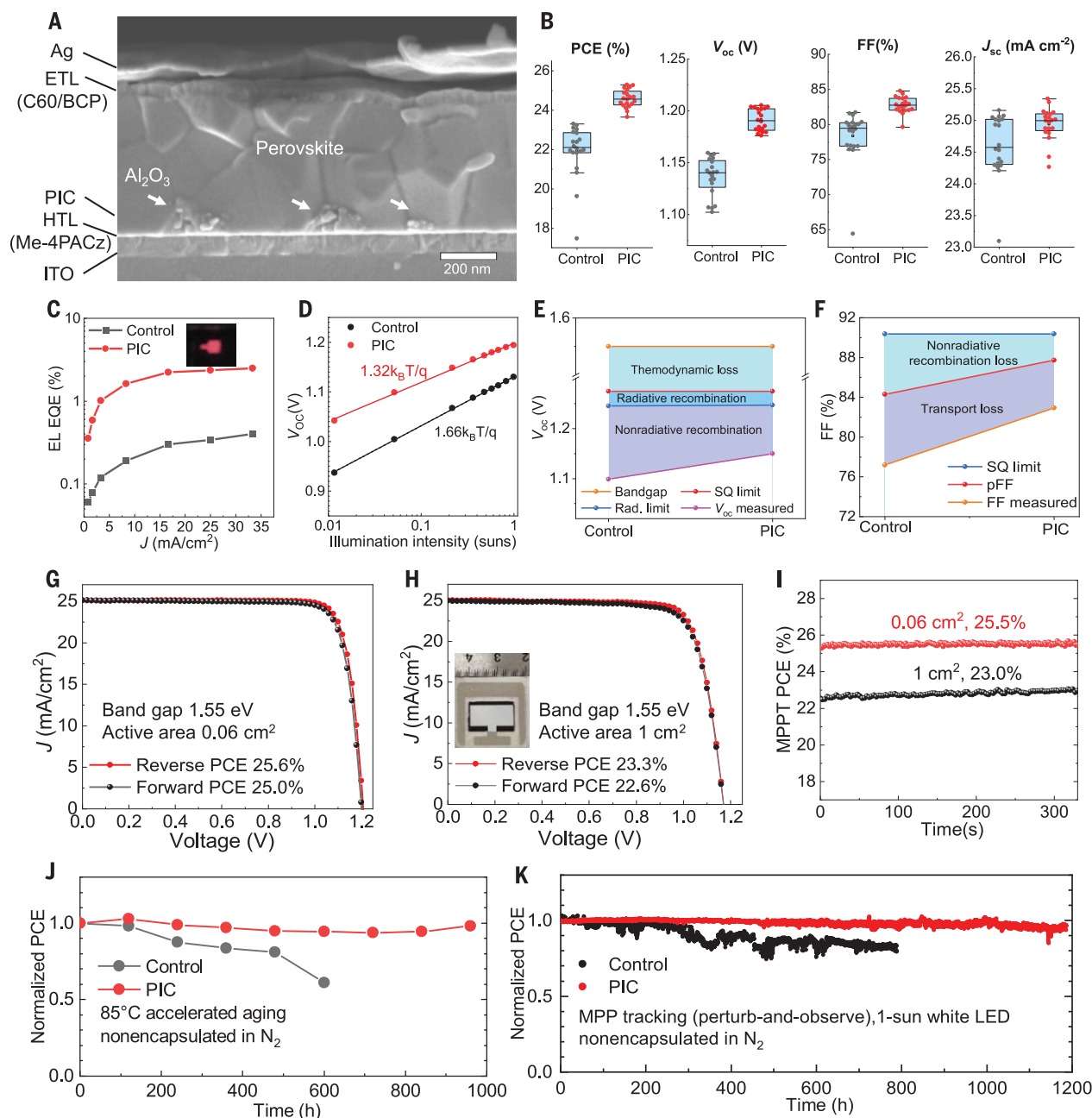


Fig. 4. PV characteristics of PIC-enhanced p-i-n devices. (A) Cross-sectional SEM image of the PIC device. The HTL is Me-4PACz coated with PIC (area coverage ~25%). Arrows indicate the Al_2O_3 islands of PIC. (B) Distribution of PCE, V_{oc} , FF, and J_{sc} of control and PIC devices. Data are shown for 20 substrates of each type made from the same perovskite precursor and processing batch. The control device uses Me-4PACz without the coating of PIC. (C) The external electroluminescence quantum efficiency (EL-EQE) measurements of control and PIC devices under different injection currents. The inset photo shows a PIC solar cell operated as an LED. (D) Suns- V_{oc} measurements to

deduce the diode ideality factor. (E and F) Detailed V_{oc} loss analysis (E) and FF loss analysis (F) of the control and PIC devices. (G and H) J - V curves of the champion PIC devices with an active area of 0.06 cm^2 (G) and 1 cm^2 (H). The perovskite absorber's band gap is 1.55 eV . (I) Stabilized PCEs of champion PIC devices determined by MPPT for 300 s. (J and K) Long-term accelerated aging test of thermal stability (continuous heating at 85°C) (J) and continuous MPPT operational stability (1-sun equivalent white LED illumination) (K) of the control and PIC devices. The devices are nonencapsulated and tested in ambient N_2 to reveal their inherent properties.

nanoplates, and its effect becomes non-ignorable at the concentration of 4 mg ml^{-1} . We also note that as the coverage increases, the local opening width can become very narrow (such as $<10 \text{ nm}$) and thus may induce non-ideal fac-

tors such as the failure of perovskite infill of narrow local openings or the dimensional-limited perovskite in the narrow channels exhibiting charge transport properties differing from the bulk (36, 37). These non-ideal factors,

from a positive viewpoint, provide valuable guidelines for further optimization of solution-processed PIC. The above PIC device model can also be more precise if we can find a way to include these non-ideal factors at high coverage.

With these studies, we henceforth used 100-nm Al_2O_3 with a concentration of 1.4 to 2 mg ml^{-1} for the optimized PIC solar cells.

For the concentration of 1.4 mg ml^{-1} with coverage of ~25%, we obtained a maximized PCE percentage improvement of ~12% (19.7 versus 22.2%) (Fig. 2, M to Q), in agreement with the above discussion that the simulator's prediction is conservative. Improvement of V_{oc} and FF could arise from factors other than just the PIC's reduction in contact area: the passivation of perovskite surface at HTL and the improved bulk quality as the perovskite grows on PIC-coated HTL. To evaluate these effects, we compared photoluminescence quantum yield (PLQY) and transient photoluminescence lifetime (TRPL) of glass/perovskite and glass/PIC/perovskite samples in the absence of HTL (Fig. 3, A to C). The sample was illuminated through the glass to be consistent with the solar cell's operational condition (19, 38). PLQY enhancement can be observed for PIC samples of concentrations from 0.7 to 2 mg ml^{-1} . Consistent with PLQY enhancement, we observed a significantly slower TRPL decay in all PIC samples with concentrations of 0.7 to 2 mg ml^{-1} , suggesting suppressed nonradiative recombination in bulk and surfaces.

We then measured the PLQY and TPPL of perovskite coated on PTAA. Consistent with other reports (6, 19), we observed that the biexponential TRPL decay of an ITO/PTAA/perovskite stack was much faster than that of a glass/perovskite stack (fig. S6) because of the additional nonradiative recombination pathways. As pointed out previously (39–42), the fast initial decay represents the combined process of charge extraction (quenching) and nonradiative recombination. The slow monoexponential decay lifetime [τ , indicated by the black dashed line in fig. S6] is mainly attributed to trap-assisted recombination in the bulk and surfaces and is among the most widely used and trusted indicators to predict the achievable V_{oc} (for details, see the materials and methods) (19, 38). Following this reasoning, we attributed the similar initial fast decays in the ITO/PTAA/perovskite stack and the ITO/PTAA/PIC/perovskite stack to the retained carrier extraction and the significantly increased τ with the use of PIC to the reduced nonradiative recombination (fig. S6). These TRPL changes agree with the V_{oc} and FF improvement observed in PIC-coated PTAA devices (Fig. 2, N to Q).

We further tested the PIC's applicability on different HTLs. We used a self-assembled monolayer of [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl] phosphonic acid (Me-4PACz) because this has been demonstrated to increase the PL lifetime and enable superior interface passivation to the PTAA/perovskite stack (6). The ITO/Me-4PACz/perovskite stack exhibited a biexponential decay consisting of a fast decay and

Table 1. PV parameters of PIC-enhanced p-i-n perovskite solar cells (band gap = 1.55 eV).

Area (cm^2)	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)	Stabilized PCE* (%)
0.06	1.208	25.08	84.37	25.56	25.50
1	1.172	25.05	79.48	23.30	23.04
0.058	1.203	25.02	82.73	24.90†	24.71†

*Maximum power point tracking for 300 s. †Certified.

an impressively long τ of ~880 ns (Fig. 3, D and E), corresponding to an achievable V_{oc} of 1.18 V (for details, see the materials and methods) (19, 38). By coating the Me-4PACz with PIC, the τ was further increased by a factor of ~4.4 (Fig. 3E). The τ of ~3800 ns in a Me-4PACz/PIC/perovskite stack corresponds to an achievable V_{oc} of 1.25 V, representing 97.3% of the S-Q limit of V_{oc} (1.285 V) for a 1.55-eV-band gap semiconductor. The increased τ in Me-4PACz/PIC/perovskite stacks was highly consistent for different PIC concentrations (0.7 to 2 mg ml^{-1}). At the same time, the TRPL and PLQY trends in Me-4PACz/PIC/perovskite stacks (fig. S7) agreed well with those observed in glass/PIC/perovskite stacks. This consistency suggests that the passivation effect of PIC and Me-4PACz can be successfully combined in the PIC device because the perovskite is in contact with both the HTL and PIC materials.

We also conducted the differential TRPL analysis to resolve the stages of charge extraction and recombination (6, 43, 44). The lifetime plateau increased from ~1 to ~4.4 μs upon the inclusion of PIC, reconfirming the substantially reduced nonradiative recombination losses in bulk and surfaces (fig. S8). The steep lifetime rise at early times was maintained in PIC samples, indicating that the local openings in PIC support the fast charge transfer.

To disentangle the bulk and surface recombination, we conducted SRV analysis by measuring TRPL as a function of perovskite thickness (Fig. 3F and fig. S9) following a double-heterostructure model (43, 45–48) as follows:

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_b} + \frac{2\text{SRV}}{d} \quad (1)$$

where τ_{eff} is the effective lifetime of TRPL decay, τ_b the bulk lifetime, and d the perovskite thickness (for details, see the materials and methods). The inclusion of PIC at HTL substantially improved the bulk lifetime from 1.2 to 6.0 μs , likely because of the hydrophilic and rough surface of Al_2O_3 clusters that profoundly modify the perovskite's growth and bulk quality, as previously reported (33, 49). This fact is evidenced by the markedly improved wetting and coverage of perovskite on the PIC-coated HTL (fig. S10). The enlarge-

ment of the perovskite's apparent grain size is not appreciable, as shown in the scanning electron microscope (SEM) top-view images and AFM morphology images in figs. S11 and S12. However, our grazing incidence wide-angle x-ray scattering (GIWAXS) data (fig. S13, A and B) showed that the PIC sample exhibited higher radially integrated intensities of perovskite (100) and (200) crystal planes relative to the control sample (fig. S13, C to E), consistent with the peak ratios observed in x-ray diffraction measurements (fig. S14). Moreover, the azimuthal integration of (100) and (200) planes showed that the PIC sample's diffraction signals were more oriented and concentrated at azimuth angles around 90° (fig. S13F), suggesting that the presence of PIC promoted its out-of-plane growth. Per previous reports (50–53), the enhancement and orientation promotion of the (100) plane have been widely linked with reduced defect density and improved carrier lifetime in bulk perovskite, in agreement with the above improvement shown in TRPL, PLQY, and SRV measurements. The SRV at the HTL interface was substantially reduced from 64.2 to 9.2 cm/s, highlighting the opportunity to approach the thermodynamic limit according to previous device simulations (46, 54).

Motivated by this success, we fabricated the p-i-n solar cells of ITO/Me-4PACz/PIC/perovskite/C60/BCP/Ag. The Al_2O_3 isolated islands (coverage of ~25% and concentration of 1.4 mg ml^{-1}) formed on the Me-4PACz layer as shown in cross-sectional SEM images (Fig. 4A). We fabricated 20 solar cells with the PIC and 20 without it in the same batch. The PIC increased the average PCE from 22.1 to 24.5% (Fig. 4B), primarily caused by an increase in V_{oc} and FF. The PIC increased the average V_{oc} from 1.13 to 1.19 V, agreeing with the above prediction from TRPL measurements (1.18 versus 1.25 V). The additional V_{oc} loss in the solar cell relative to the prediction can be linked to the non-idealities of fabrication and additional loss at the ETL contact (6, 19, 55).

We conducted a quantitative analysis of the V_{oc} loss following the detailed balance theory (for details, see the materials and methods) (56, 57). The external quantum efficiency of electroluminescence of the PIC device (fig. S15)

increased by a factor of ~ 7 relative to the control at the injection level close to J_{sc} (Fig. 4C), which is consistent with the PLQY and SRV improvements. The nonradiative recombination loss was reduced by ~ 50 mV, representing the main contributor to the higher V_{oc} in PIC devices (Fig. 4E). By contrast, the improvement (~ 2 mV) in radiative recombination loss was secondary, as revealed by the high-sensitivity external quantum efficiency near the band edge (fig. S16).

To quantitatively analyze the enhanced FF in PIC devices (from the average of 78.5 to 82.3%) (Fig. 4B), we used the Suns- V_{oc} method, which involves measuring J - V curves as a function of light intensity (for details, see the materials and methods) (58, 59). The deduced ideality factor was improved from 1.66 to 1.32 (Fig. 4D), in good agreement with nonradiative recombination suppression and FF enhancement. The FF enhancement in PIC devices has considerable contributions from reduced transport loss (4.78 versus 7.11%) and reduced nonradiative recombination loss (2.65 versus 6.05%) (Fig. 4F). The PIC enabled the pseudo-FF, which is the FF in the absence of transport loss, to be 87.7% (fig. S17), which is $\sim 97.1\%$ of the thermodynamic limit (S-Q limit). By contrast, the pseudo-FF for the bare Me-4PACz was only 84.3%. This FF analysis highlights how the PIC overcomes the V_{oc} -FF trade-off by forming a practical, through-the-opening transport through the perovskite infill.

The champion PIC device showed a PCE of 25.6% ($V_{oc} = 1.208$ V, $J_{sc} = 25.08$ mA cm $^{-2}$, and FF = 84.37%) with negligible hysteresis (Fig. 4G). The stabilized PCE was 25.5% after a maximum-power-point tracking for 300 s (Fig. 4I). This efficiency is among the highest ever recorded for perovskite single-junction cells (table S1). The measured V_{oc} and FF are 94 and 93.5% of their S-Q limit for a 1.55 eV solar cell and $\sim 97\%$ of their upper limits predicted from TRPL lifetime (1.25 V) and pseudo-FF (87.7%), respectively. As previously reported (1), the product $V_{oc} \times$ FF relative to the S-Q limit allows quantifying the V_{oc} -FF trade-off without the influence of different band gaps. Our PIC device can deliver a $V_{oc} \times$ FF of 87.9% of the S-Q limit, the highest for a p-i-n device to date and rivaling the level (87.3%) in the record n-i-p device (table S1 and fig. S18).

The certification of one PIC device measured by an independently accredited testing center (the Chinese National PV industry Measurement and Testing Center) gave a PCE of 24.9% ($V_{oc} = 1.203$ V, $J_{sc} = 25.02$ mA cm $^{-2}$, and FF = 82.73%) (fig. S19) and a stabilized PCE of 24.71% after maximum-power-point tracking (MPPT) for 300 s (fig. S20). This confirmed value is the highest certified efficiency among all p-i-n devices (table S1). The J_{sc} measured in our laboratory highly matched the certified J_{sc} within an error of $<0.3\%$ and also matched the

J_{sc} (24.4 mA cm $^{-2}$) calculated by integrating the EQE spectrum (fig. S21).

From the device manufacturing perspective, the standard deviation in PCE was reduced from 1.37 to 0.45% by using the PIC (Fig. 4B). The improved homogeneity and yield of perovskite films deposited on the rough and hydrophilic PIC surface allowed us to realize enlarged cells with fewer shunting pathways. The PIC cells with an aperture area of 1 cm 2 reached a PCE of 23.3% ($V_{oc} = 1.17$ V, $J_{sc} = 25.05$ mA cm $^{-2}$, and FF = 79.48%) (Fig. 4H), among the highest values for all perovskite cells with active area ≥ 1 cm 2 (table S2).

Stability measurements

We examined the long-term stability of PIC devices under accelerated lifetime conditions of continuous heating and illumination following the ISOS consensus protocol (for details, see the materials and methods) (60). An SnO $_2$ layer (15 to 20 nm) made by atomic layer deposition was used to replace the BCP layer for the long-term stability test without a trade-off of PCE. We tested unencapsulated devices in an N $_2$ atmosphere to reveal their intrinsic properties and to avoid encapsulation modification. The PIC devices retained 98% of the initial PCE after accelerated aging at 85°C for 1000 hours, whereas the control device exhibited $\sim 20\%$ PCE loss (T_{80} lifetime) at ~ 400 hours (Fig. 4J). Moreover, the PIC devices exhibited minor degradation of $<2\%$ after 1000 hours of continuous operation at MPPT (perturb-and-observe tracking) under 1-sun illumination (chamber temperature ~ 35 to 40° C) (Fig. 4K). The control devices showed no degradation for ~ 200 hours and then a 20% loss of PCE at ~ 800 hours. We noted that our p-i-n devices exhibited no burn-in degradation but a slightly increased PCE during the initial stage, which may have been caused by light-induced defect healing of perovskite, as revealed in previous reports (61, 62). This impressive operational stability in PIC devices is consistent with the observed improvement in film wetting, passivation, and carrier transport. It has been revealed that better passivation paired with faster carrier extraction at the perovskite/HTL interface can effectively improve stability because of less carrier accumulation and suppressed phase segregation (6, 63).

Outlook

In light of the successful application of double-sided (both n- and p-type) local contacts in silicon cells, we attempted the PIC device's extension by additionally spin-coating Al $_2$ O $_3$ on the perovskite surface but did not gain any further improvement in V_{oc} and FF (fig. S22A). We speculate that Al $_2$ O $_3$ clustering and growth on the perovskite rough surface may differ from that on the HTL flat surface, and the local contact structure at the rear ETL differs

from that of the front PIC (see the schematic diagrams in fig. S22, B to E), suggesting that the extension of PIC on the top surface requires appropriate modification. It is also worth noting that the established silicon solar cells' local contacts are typical of a coverage of $>90\%$. Consistently, our simulations (fig. S4) showed that the PCE could be raised by another $\sim 2\%$ if a way could be found to fabricate a nearly ideal PIC structure with coverage increased from 25 to $>80\%$.

The PIC concept can also be generalized to other perovskite compositions. In FA $_{0.95}$ MA $_{0.05}$ Pb(I $_{0.92}$ Br $_{0.08}$) $_3$, a Cs-free composition with a band gap of 1.57 eV, we improved the p-i-n cells' PCE from 22.90 to 25.13% by using PIC, featured with high V_{oc} of 1.2 V and FF of 84.14% (fig. S23). Moreover, we also observed considerable PCE, V_{oc} , and FF improvement in 1.65-eV perovskite cells, which are of high interest for tandem applications (fig. S24). These two demonstrations were not optimized. For instance, the PIC parameters, such as the opening spacing, should be tuned to match the photo-carrier diffusion length of a specific perovskite.

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SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S24

Tables S1 to S4

References (64–85)

Simulation Model File

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